Correlation Functions in a Finite-Size Kitaev Model

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Abstract

In this note, we discuss how to compute multi-spin correlation functions in a finite-size Kitaev model, which only contains four distinct plaquettes. These correlation functions are crucial in constructing the density matrix of the Kitaev ground state.

1 Foundations

1.1 Correlation Functions

As mentioned in our work [1], at the ground state, the multi-spin correlation function of the Kitaev model is finite if and only if it can be written in the form of a string. Therefore, we only need to discuss the following form of the correlation function

$$\langle \prod_{\langle ij\rangle \in \mathfrak{D}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \rangle$$

This kind of correlation function can be exactly evaluated in the Majorana representation. Substituting the transformation

$$\sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \to i b_i^{\alpha_{ij}} c_i i b_j^{\alpha_{ij}} c_j = \hat{u}_{ij} i c_j c_i$$

into the correlation function and taking the trivial gauge, we have

$$\langle \prod_{\langle ij\rangle \in \mathfrak{D}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \rangle = \langle \prod_{\langle ij\rangle \in \mathfrak{D}} i c_j c_i \rangle_M$$

1.2 Minimal Size for Non-trivial Topology

For numerical efficiency, we tend to choose the smallest possible system. However, the system must still be sufficiently large such that the topological order still exists. A requirement for this is that the ground state manifold should possess four independent topological sectors generated by global Wilson loops if the geometry of the system is a torus.

We start from the Wen plaquette model. There are two requirements for a Wen plaquette system to possess four topologically degenerate ground states:

- The lattice must contain an even number of rows or columns in both the horizontal and vertical directions.
- A global Wilson loop cannot be generated or destroyed by local plaquette operators.

The physical meaning of the first requirement is to preserve the checkerboard pattern of the lattice. If there are adjacent plaquettes (cells) that are of the same type (black or white, in a checkerboard sense), then the two types of plaquettes become identical, and the original two different Wilson loops become the same. This breakdown of the checkerboard pattern reduces the original fourfold topological ground state degeneracy down to twofold, which is not adequate to describe the physics of a toric code system.

According to the two requirements, **the smallest possible topologically non-trivial Wen plaquette model contains eight ($8 = 2 \times 4$) distinct plaquettes and eight sites**. A 2×2 checkerboard is not adequate for the system to have four topological sectors since it violates the second requirement[1].

[1] The *global Wilson loop* is identical to a plaquette operator

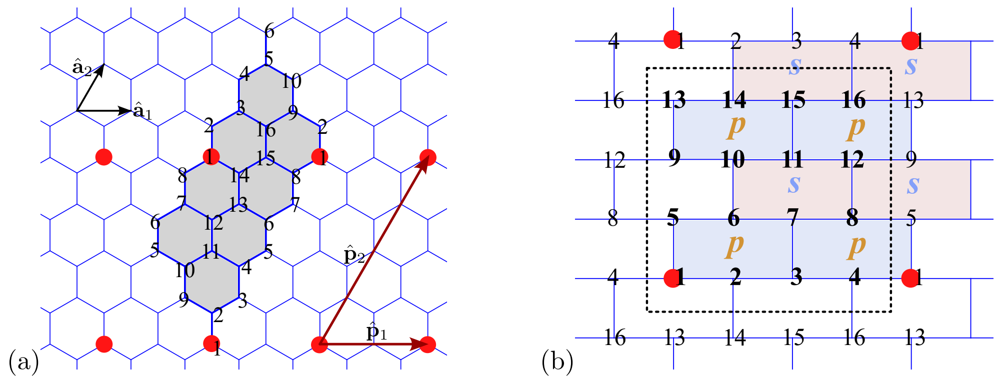

Figure 1: (a) The smallest topologically non-trivial system consists of 8 unit cells, which is equivalent to 16 spins. The two vectors that define the periodic boundary condition are $\hat{\mathbf{p}}_1 = 2\hat{\mathbf{a}}_1$ and $\hat{\mathbf{p}}_2 = 4\hat{\mathbf{a}}_2$, respectively. (b) For numerical simplicity, the honeycomb lattice is rearranged into a brick-wall geometry, and the spins are relabeled according to their Euclidean coordinates in the lattice. In both figures, red dots indicate identical lattice sites.

2 Finite-Size Correlation

We only have to focus on the two-point correlation functions. Here we set that the system satisfies the **periodic boundary condition**, which is characterized by two independent vectors, $\hat{\mathbf{p}}_1 = 2\hat{\mathbf{a}}_1$ and $\hat{\mathbf{p}}_2 = 4\hat{\mathbf{a}}_2$, as illustrated in Fig.1.a. The geometry adopted here follows the one illustrated in Ref. [2]

2.1 Momentum States in Finite Systems

For slave fermions in the parton construction, the boundary condition of the system could be periodic (PBC) or anti-periodic (APBC), depending on which topological sector we choose [3]. Here, we only consider the PBC in both the horizontal and vertical directions.

$$f_{\sigma\mu\mathbf{x}} = f_{\sigma\mu,(\mathbf{x}+\hat{\mathbf{p}}_1)} = f_{\sigma\mu,(\mathbf{x}+\hat{\mathbf{p}}_2)} \tag{1}$$

To determine the allowed momentum states, we first Fourier transform the fermions as follows

$$f_{\sigma\mu\mathbf{x}} = \frac{1}{\sqrt{N/2}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{x}} f_{\sigma\mu\mathbf{k}}$$

With this expression, Eq. (1) implies that the allowed momentum $\mathbf{k}$ must satisfy

$$\left\{ \begin{array}{lll} \mathbf{k}\cdot\hat{\mathbf{p}}_1 = 2n_1\pi & ; & n_1 = 0, 1, \cdots, N_h - 1 \\ \mathbf{k}\cdot\hat{\mathbf{p}}_2 = 2n_2\pi & ; & n_2 = 0, 1, \cdots, N_v - 1 \end{array} \right. \tag{2}$$

In our case illustrated in Fig. 1, $N_h = 2$ and $N_v = 4$, representing the number of unit cells included in the system in the horizontal and vertical directions, respectively. Consider the unit vectors $\hat{\mathbf{a}}_1 = \hat{\mathbf{e}}_x$ and $\hat{\mathbf{a}}_2 = \frac{1}{2}\hat{\mathbf{e}}_x + \frac{\sqrt{3}}{2}\hat{\mathbf{e}}_y$. We can define the unit vectors in the momentum space as

$$\begin{cases} \hat{\mathbf{b}}_1 = 2\pi\hat{\mathbf{e}}_y \\ \hat{\mathbf{b}}_2 = \sqrt{3}\pi\hat{\mathbf{e}}_x - \pi\hat{\mathbf{e}}_y \end{cases}$$

such that $\hat{\mathbf{a}}_i \cdot \hat{\mathbf{b}}_j = 2\pi\delta_{ij}$. With these basis vectors, the allowed momentum states are

$$\mathbf{k} = \frac{n_1}{2}\hat{\mathbf{b}}_1 + \frac{n_2}{4}\hat{\mathbf{b}}_2 \quad ; \quad n_1 = 0, 1 \quad , \quad n_2 = 0, 1, 2, 3 \tag{3}$$

2.2 Two-Points Correlation Functions

The correlation functions can be calculated more easily in the normal fermion language. As defined in the appendix, the c-Majorana fermions can be written in terms of γ-fermions,

$$\begin{cases} c_{\mathbf{x}A} = \gamma_{\mathbf{x}}^{\dagger} + \gamma_{\mathbf{x}} = \dfrac{1}{\sqrt{N/2}} \sum_{\mathbf{k}} \left(\gamma_k^{\dagger} e^{-ik\mathbf{x}} + \gamma_{\mathbf{k}} e^{ik\mathbf{x}} \right) \\ c_{\mathbf{x}B} = i \left(\gamma_{\mathbf{x}}^{\dagger} - \gamma_{\mathbf{x}} \right) = \dfrac{i}{\sqrt{N/2}} \sum_{\mathbf{k}} \left(\gamma_k^{\dagger} e^{-ik\mathbf{x}} - \gamma_{\mathbf{k}} e^{ik\mathbf{x}} \right) \end{cases}$$

And the ground state is

$$|\psi\rangle = A \exp\left(\sum_{k \in \mathrm{HBZ}} g_k \gamma_k^{\dagger} \gamma_{-k}^{\dagger} \right)$$

where

$$A = \prod_{k \in \mathrm{HBZ}} \left(1 + |g_k|^2 \right)^{-1/2}$$

is the normalization constant. Similar to Numbu-Gor'kov Green's function, there are two types of fermion pairing, $AA(BB)$ and AB. Both types of correlation functions can be simplified to the problem of solving the γ-fermion correlation function as we will see later. The two γ-fermion correlation functions we have to know are $\langle \gamma_{\mathbf{k}} \gamma_{k'} \rangle$ and $\langle \gamma_k^{\dagger} \gamma_{k'} \rangle$. Their analytical forms are given as follows

$$\begin{cases} \langle \gamma_k^{\dagger} \gamma_{k'} \rangle = \dfrac{|g_k|^2}{1 + |g_k|^2} \delta_{kk'} = v_k^2 \delta_{kk'} = \dfrac{1}{2} \left(1 - \dfrac{R_{\mathbf{k}}}{|s_{\mathbf{k}}|} \right) \delta_{kk'} \\ \langle \gamma_{\mathbf{k}} \gamma_{k'} \rangle = \dfrac{g_k}{1 + |g_k|^2} \delta_{k,-k'} = i \operatorname{sgn}(I_k)\, u_k v_k \delta_{k,-k'} = i \dfrac{I_k}{2|s_{\mathbf{k}}|} \delta_{k,-k'} \end{cases}$$

and the fermionic commutation relations can give other two types of correlation functions

$$\begin{cases} \langle \gamma_{\mathbf{k}} \gamma_{k'}^{\dagger} \rangle = \delta_{kk'} - \langle \gamma_{k'}^{\dagger} \gamma_{\mathbf{k}} \rangle = u_k^2 \delta_{kk'} = \dfrac{1}{2} \left(1 + \dfrac{R_{\mathbf{k}}}{|s_{\mathbf{k}}|} \right) \delta_{kk'} \\ \langle \gamma_{\mathbf{k}}^{\dagger} \gamma_{k'}^{\dagger} \rangle = \langle \gamma_{k'} \gamma_{\mathbf{k}} \rangle^* = -i \operatorname{sgn}(I_{k'}) u_{k'} v_{k'} \delta_{k',-k} = i \operatorname{sgn}(I_k) u_k v_k \delta_{k,-k'} = \langle \gamma_{\mathbf{k}} \gamma_{k'} \rangle \end{cases}$$

With these relations, we can start to compute the c-Majorana correlation functions. For AA-pairings, the correlation function is,

$$\begin{aligned} \langle c_{\mathbf{x}A} c_{\mathbf{x}'A} \rangle &= \frac{1}{N/2} \sum_{\mathbf{k}} \left(\langle \gamma_{\mathbf{k}} \gamma_{-\mathbf{k}} \rangle + \langle \gamma_{-\mathbf{k}}^{\dagger} \gamma_k^{\dagger} \rangle + \langle \gamma_{\mathbf{k}} \gamma_k^{\dagger} \rangle + \langle \gamma_{-\mathbf{k}}^{\dagger} \gamma_{-\mathbf{k}} \rangle \right) e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}')} \\ &= \frac{1}{N/2} \sum_{\mathbf{k}} \left(1 - \frac{R_{\mathbf{k}}}{|s_{\mathbf{k}}|} \right) e^{i\mathbf{k} \cdot (\mathbf{x}' - \mathbf{x})} \\ &= \frac{1}{N/2} \sum_{\mathbf{k}} \left(1 - \frac{R_{\mathbf{k}}}{|s_{\mathbf{k}}|} \right) \cos\left(\mathbf{k} \cdot (\mathbf{x}' - \mathbf{x}) \right) \in \mathbb{R} \end{aligned}$$

The imaginary part is an odd function in the momentum space, hence it vanishes after the integration. However, this implies that the Majorana correlation function is a real number, which is impossible since

$$\langle c_i c_j \rangle^* = \langle c_j c_i \rangle = -\langle c_i c_j \rangle$$

which must be an imaginary number unless $i = j$. So we conclude that

$$\langle c_{\mathbf{x}A} c_{\mathbf{x}'A} \rangle = \langle c_{\mathbf{x}B} c_{\mathbf{x}'B} \rangle = \delta_{\mathbf{x}\mathbf{x}'} \tag{4}$$

As for AB-pairings, a similar calculation gives

$$\begin{aligned} \langle c_{\mathbf{x}A} c_{\mathbf{x}'B} \rangle &= \frac{i}{N/2} \sum_{\mathbf{k}} \left(\langle -\gamma_{\mathbf{k}} \gamma_{-\mathbf{k}} \rangle + \langle \gamma_{-\mathbf{k}}^{\dagger} \gamma_k^{\dagger} \rangle + \langle \gamma_{\mathbf{k}} \gamma_k^{\dagger} \rangle - \langle \gamma_{-\mathbf{k}}^{\dagger} \gamma_{-\mathbf{k}} \rangle \right) e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}')} \\ &= \frac{i}{N/2} \sum_{\mathbf{k}} \left(\frac{R_{\mathbf{k}}}{|s_{\mathbf{k}}|} - i \frac{I_k}{|s_{\mathbf{k}}|} \right) e^{i\mathbf{k} \cdot (\mathbf{x}' - \mathbf{x})} = \frac{i}{N/2} \sum_{\mathbf{k}} \frac{s_{\mathbf{k}}^*}{|s_{\mathbf{k}}|} e^{i\mathbf{k} \cdot (\mathbf{x}' - \mathbf{x})} \end{aligned}$$

This result is pure imaginary, and can be further simplified,

$$\langle c_{\mathbf{x}A} c_{\mathbf{x}'B} \rangle = \frac{i}{N/2} \sum_{\mathbf{k}} \cos\left[\mathbf{k} \cdot (\mathbf{x}' - \mathbf{x}) - \theta_{\mathbf{k}}\right] = \frac{i}{N/2} \sum_{\mathbf{k}} \frac{1}{|s_{\mathbf{k}}|} \left[R_{\mathbf{k}} \cos\left(\mathbf{k} \cdot \delta\mathbf{x}\right) + I_{\mathbf{k}} \sin\left(\mathbf{k} \cdot \delta\mathbf{x}\right)\right] \tag{5}$$

where $s_{\mathbf{k}} = |s_{\mathbf{k}}| e^{i\theta_{\mathbf{k}}}$. Notice that the above formulation is general and can be used directly in the case of finite systems.

2.3 String Correlation Functions

A Exactly Solving for the Ground State

According to equation (29) of Ref. [4], we can diagonalize the Kitaev Hamiltonian in the momentum space if the gauge is fixed to be $u_{ij} = +1 \forall\, (ij)$,

$$\hat{H} = \sum_{\mathbf{k}} \psi_k^\dagger U(k) \psi_k = \frac{1}{2} \sum_{\mathbf{k}} \left[i s_{\mathbf{k}} d_{k,A}^\dagger d_{k,B} - i s_{\mathbf{k}}^* d_{k,B}^\dagger d_{k,A} \right]$$

where

$$\psi_k \equiv \frac{1}{\sqrt{2N}} \sum_{\mu} e^{-ik\cdot r_\mu} \begin{pmatrix} c_{\mu,A} \\ c_{\mu,B} \end{pmatrix} = \begin{pmatrix} d_{k,A} \\ d_{k,B} \end{pmatrix}$$

$$U(k) = \frac{1}{2} \begin{pmatrix} 0 & i s_{\mathbf{k}} \\ -i s_{\mathbf{k}}^* & 0 \end{pmatrix} \quad , \quad s_{\mathbf{k}} = 2\left(J_x e^{ik\cdot(\mathbf{a}_2 - \mathbf{a}_1)} + J_y e^{ik\cdot \mathbf{a}_2} + J_z \right)$$

Note that the index μ above is the Bravais lattice point, and A,B represent the sublattice. If we further define a normal fermion operator,

$$\gamma_\mu \equiv \frac{1}{2} \left(c_{\mu,A} + i c_{\mu,B} \right) \tag{6}$$

then its Fourier transform would be

$$\gamma_k = \frac{1}{\sqrt{2N}} \sum_{\mu} \gamma_\mu e^{-ik\cdot r_\mu} = \frac{1}{2} \left(d_{k,A} + i d_{k,B} \right)$$

In this way, the operator d can be expressed in terms of γ fermion,

$$\begin{cases} d_{k,A} = \gamma_{-\mathbf{k}}^\dagger + \gamma_k \\ d_{k,B} = i\left(\gamma_{-\mathbf{k}}^\dagger - \gamma_k \right) \end{cases}$$

Using the γ operators to rewrite the Hamiltonian gives

$$\hat{H} = \sum_{\mathbf{k}} \left[R_{\mathbf{k}} \cdot \left(\gamma_{\mathbf{k}}^\dagger \gamma_{\mathbf{k}} - \gamma_{-\mathbf{k}} \gamma_{-\mathbf{k}}^\dagger \right) - i I_{\mathbf{k}} \cdot \left(\gamma_k^\dagger \gamma_{-\mathbf{k}}^\dagger - \gamma_{-\mathbf{k}} \gamma_k \right) \right] = \sum_{\mathbf{k}} \phi_k^\dagger \mathbf{H}(k) \phi_k$$

where $R_{\mathbf{k}} \equiv \mathrm{Re}\{s_{\mathbf{k}}\}$ and $I_k \equiv \mathrm{Im}\{s_{\mathbf{k}}\}$, and

$$\mathbf{H}(k) \equiv \begin{pmatrix} R_{\mathbf{k}} & -i I_{\mathbf{k}} \\ i I_k & -R_{\mathbf{k}} \end{pmatrix} \quad , \quad \phi_k \equiv \begin{pmatrix} \gamma_{\mathbf{k}} \\ \gamma_{-\mathbf{k}}^\dagger \end{pmatrix}$$

Since

$$\mathbf{H}(k)^2 = \left(R_{\mathbf{k}}^2 + I_k^2 \right) \mathbb{1} = |s_{\mathbf{k}}|^2 \mathbb{1}$$

we can see that the eigenvalues are $\epsilon_k = \pm |s_{\mathbf{k}}|$, which is consistent with Kitaev's result (equation (32) in Ref.[**?**]). To diagonalize the matrix $\mathbf{H}$ we have to apply the Bogoliubov transformation,

$$U = \frac{1}{\sqrt{2\,|s_{\mathbf{k}}|}} \begin{pmatrix} \mathrm{sgn}(I_k)\sqrt{|s_{\mathbf{k}}| + R_{\mathbf{k}}} & -i\sqrt{|s_{\mathbf{k}}| - R_{\mathbf{k}}} \\ i\sqrt{|s_{\mathbf{k}}| - R_{\mathbf{k}}} & -\mathrm{sgn}(I_k)\sqrt{|s_{\mathbf{k}}| + R_{\mathbf{k}}} \end{pmatrix} = \begin{pmatrix} \mathrm{sgn}(I_k) u_k & -i v_k \\ i v_k & -\mathrm{sgn}(I_k) u_k \end{pmatrix} = v_k \sigma^y + \mathrm{sgn}(I_k) u_k \sigma^z$$

such that[2]

$$U\mathbf{H}U^\dagger = \begin{pmatrix} |s_{\mathbf{k}}| & 0 \\ 0 & -|s_{\mathbf{k}}| \end{pmatrix}$$

Therefore, the excitation that diagonalize the Hamiltonian is

$$\begin{pmatrix} a_k \\ a^\dagger_{-k} \end{pmatrix} = U \begin{pmatrix} \gamma_{\mathbf{k}} \\ \gamma^\dagger_{-k} \end{pmatrix} = \begin{pmatrix} \mathrm{sgn}(I_k) u_k \gamma_{\mathbf{k}} - i v_k \gamma^\dagger_{-k} \\ i v_k \gamma_{\mathbf{k}} - \mathrm{sgn}(I_k) u_k \gamma^\dagger_{-k} \end{pmatrix} \tag{7}$$

As a result, the Kitaev ground state is the vacuum state of the Bogoliubov fermion, i.e., $a_k|M\rangle = 0$, which means $|M\rangle$ can be written as a BCS ground state just as we met in the BCS theory. The ground state can be written as the coherent state[3] of the γ-fermion pair,

$$\begin{aligned} |M\rangle &= A\,[g] \exp\left(\sum_{\mathbf{k}} g_k \gamma^\dagger_k \gamma^\dagger_{-k}\right) |\mathrm{vac}_\gamma\rangle \\ &= A\,[g] \exp\left(\sum_{\mu\nu} g_{\mu\nu} \gamma^\dagger_\mu \gamma^\dagger_\nu\right) |\mathrm{vac}_\gamma\rangle \\ &= A\,[g] \left[\mathbb{1} + \sum_{\mu\nu} g_{\mu\nu} \gamma^\dagger_\mu \gamma^\dagger_\nu + \frac{1}{2!} \sum_{\mu\nu\kappa\lambda} g_{\mu\nu} g_{\kappa\lambda} \gamma^\dagger_\mu \gamma^\dagger_\nu \gamma^\dagger_\kappa \gamma^\dagger_\lambda + \cdots \right] |\mathrm{vac}_\gamma\rangle \end{aligned}$$

where $|\mathrm{vac}_\gamma\rangle$ is the vacuum state of the γ-fermions, and $g_{\mu\nu} = \frac{1}{2N}\sum_{\mathbf{k}} g_k e^{-ik\cdot(r_\mu - r_\nu)}$.

B Sum Rules for Correlation Functions

$$\rho = \frac{1}{2^{N-N_p}} \mathcal{P}^{(p)} \sum_{[\mathfrak{D}]} \mathfrak{C}_{\mathfrak{D}} \Sigma_{\mathfrak{D}}$$

$$\mathcal{P}^{(p)} = \left(\frac{\mathbb{1} + \tilde{W}_v}{2}\right)\left(\frac{\mathbb{1} + \tilde{W}_h}{2}\right) \prod_p \left(\frac{\mathbb{1} + W_p}{2}\right)$$

$$\begin{aligned} \rho^2 &= \frac{1}{2^{2N-2N_p}} \mathcal{P}^{(p)} \sum_{[\mathfrak{D}]} \sum_{[\mathfrak{D}']} \mathfrak{C}_{\mathfrak{D}} \mathfrak{C}_{\mathfrak{D}'} \varphi(\mathfrak{D}, \mathfrak{D}') \Sigma_{\mathfrak{D} \ominus \mathfrak{D}'} \\ &= \frac{1}{2^{2N-2N_p}} \mathcal{P}^{(p)} \sum_{[\mathfrak{D}]} \sum_{[\mathfrak{D}']} \mathfrak{C}_{\mathfrak{D}} \mathfrak{C}_{\mathfrak{D}'} \frac{1}{2} \left[\varphi(\mathfrak{D}, \mathfrak{D}') + \varphi(\mathfrak{D}', \mathfrak{D})\right] \Sigma_{\mathfrak{D} \ominus \mathfrak{D}'} \\ &= \frac{1}{2^{2N-2N_p}} \mathcal{P}^{(p)} \sum_{[\mathfrak{D}]} \sum_{[\mathfrak{D}']} \mathfrak{C}_{\mathfrak{D}} \mathfrak{C}_{\mathfrak{D}'} \vartheta(\mathfrak{D}, \mathfrak{D}') \Sigma_{\mathfrak{D} \ominus \mathfrak{D}'} \\ &= \frac{1}{2^{2N-2N_p}} \mathcal{P}^{(p)} \sum_{[\mathfrak{D}^c]} \sum_{[\mathfrak{D}']} \mathfrak{C}_{\mathfrak{D}^c \ominus \mathfrak{D}'} \mathfrak{C}_{\mathfrak{D}'} \vartheta(\mathfrak{D}^c \ominus \mathfrak{D}', \mathfrak{D}') \Sigma_{\mathfrak{D}^c} \\ &= \frac{1}{2^{N-N_p}} \mathcal{P}^{(p)} \sum_{[\mathfrak{D}]} \left(\frac{1}{2^{N-N_p}} \sum_{[\mathfrak{D}']} \mathfrak{C}_{\mathfrak{D} \ominus \mathfrak{D}'} \mathfrak{C}_{\mathfrak{D}'} \eta_{\mathfrak{D}'} \vartheta(\mathfrak{D}, \mathfrak{D}') \right) \Sigma_{\mathfrak{D}} \end{aligned}$$

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[2]Those parameters in the matrix U have the following properties: **(i)** $u_k = u_{-k}$, **(ii)** $v_k = v_{-k}$, and **(iii)** $\mathrm{sgn}(I_k) = -\mathrm{sgn}(I_{-k})$

[3]The coefficient g_k can be represented in terms of u_k and v_k. As I have mentioned, the BCS-like ground state is expected to be the vacuum state of a_k fermions for all possible k, thus, we have

$$\left(\mathrm{sgn}(I_k) u_k \gamma_{\mathbf{k}} - i v_k \gamma^\dagger_{-\mathbf{k}}\right) \prod_k \left(\mathbb{1} + g_k \gamma^\dagger_{\mathbf{k}} \gamma^\dagger_{-\mathbf{k}}\right) |\mathrm{vac}_\gamma\rangle = 0$$

Based on this, we obtain that

$$\mathrm{sgn}(I_k) u_k g_k - i v_k = 0$$

therefore

$$g_k = \mathrm{sgn}(I_k) i \frac{v_k}{u_k} = \mathrm{sgn}(I_k) i \sqrt{\frac{|s_{\mathbf{k}}| - R_{\mathbf{k}}}{|s_{\mathbf{k}}| + R_{\mathbf{k}}}} = i \frac{|s_{\mathbf{k}}| - R_{\mathbf{k}}}{I_k} \tag{8}$$

$$\frac{1}{2^{N-N_p}} \sum_{[\mathfrak{D}']} \vartheta(\mathfrak{D}, \mathfrak{D}') \mathfrak{C}_{\mathfrak{D} \ominus \mathfrak{D}'} \mathfrak{C}_{\mathfrak{D}'} = \mathfrak{C}_{\mathfrak{D}} \tag{9}$$

$$\frac{1}{2^{N-N_p}} \sum_{[\mathfrak{D}']} \mathfrak{C}^2_{\mathfrak{D}'} = 1 \tag{10}$$

notice that $\eta_{\mathfrak{D}} = 1$ if $[\mathfrak{D}] \in \mathbf{G}$

$$\vartheta(\mathfrak{D}, \mathfrak{D}') = \frac{1}{2} \left[1 + \eta_{\mathfrak{D}} \eta_{\mathfrak{D}'} \eta_{\mathfrak{D}' \ominus \mathfrak{D}}\right] \varphi(\mathfrak{D}, \mathfrak{D}') = \vartheta(\mathfrak{D}', \mathfrak{D}) = 0 \text{ or } \pm 1$$

$$\varphi(\mathfrak{D} \ominus \mathfrak{D}', \mathfrak{D}') = \eta_{\mathfrak{D}} \eta_{\mathfrak{D} \ominus \mathfrak{D}'} \varphi(\mathfrak{D}', \mathfrak{D}) = \eta_{\mathfrak{D}'} \varphi(\mathfrak{D}, \mathfrak{D}')$$

$$\vartheta(\mathfrak{D} \ominus \mathfrak{D}', \mathfrak{D}') = \frac{1}{2} \left[1 + \eta_{\mathfrak{D} \ominus \mathfrak{D}'} \eta_{\mathfrak{D}'} \eta_{\mathfrak{D}}\right] \varphi(\mathfrak{D} \ominus \mathfrak{D}', \mathfrak{D}') = \eta_{\mathfrak{D}'} \vartheta(\mathfrak{D}, \mathfrak{D}')$$

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